



Group 18 - Influence of Pole and Zero Location on the Step Response

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1 System Definition

The objective of this exercise is to investigate how the location of poles and zeros in the z -plane affects the step response of a discrete-time system. A nominal second-order system was first defined, after which the zero and pole locations were varied separately. The steady-state gain was normalized to one in every case so that the observed differences correspond mainly to changes in the transient dynamics.

The transfer function considered was

$$H(z) = K \frac{z - z_0}{z^2 - 2r \cos(\theta)z + r^2}, \quad (1)$$

with complex-conjugate poles

$$p_{1,2} = re^{\pm j\theta}. \quad (2)$$

An equivalent continuous-time damping ratio of $\zeta = 0.5$ was selected, together with a nominal pole angle of

$$\theta_{\text{nom}} = 45^\circ.$$

For an underdamped continuous-time pole pair,

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}. \quad (3)$$

Using the discrete-time mapping $z = e^{sT_s}$ gives

$$z_{1,2} = e^{-\zeta\omega_n T_s} e^{\pm j\omega_n \sqrt{1-\zeta^2} T_s}. \quad (4)$$

Comparing this expression with the polar representation

$$z_{1,2} = re^{\pm j\theta}$$

gives

$$r = e^{-\zeta\omega_n T_s}, \quad \theta = \omega_n \sqrt{1-\zeta^2} T_s.$$

Eliminating $\omega_n T_s$ results in

$$\boxed{r = \exp\left(-\frac{\zeta\theta}{\sqrt{1-\zeta^2}}\right)}. \quad (5)$$

For $\zeta = 0.5$ and $\theta = 45^\circ$, this gives

$$r_{\text{nom}} \approx 0.635, \quad p_{1,2} \approx 0.449 \pm j0.449.$$

The corresponding nominal zero from Eq. (4.104) is

$$z_{0,\text{nom}} = r_{\text{nom}} \cos(\theta_{\text{nom}}) \approx 0.449.$$

1.1 Steady-State Gain Normalization

Changing the pole or zero location generally also changes the steady-state gain. To compare the transient behaviour independently of this effect, the gain K was adjusted for every configuration such that

$$H(1) = 1. \quad (6)$$

For a unit-step input,

$$U(z) = \frac{z}{z-1},$$

and the discrete-time final value theorem gives

$$y(\infty) = \lim_{z \rightarrow 1} (1 - z^{-1})Y(z).$$

Since $Y(z) = H(z)U(z)$,

$$\begin{aligned} y(\infty) &= \lim_{z \rightarrow 1} \left(\frac{z-1}{z} \right) H(z) \frac{z}{z-1} \\ &= H(1). \end{aligned}$$

Therefore, imposing $H(1) = 1$ ensures that all tested systems have the same steady-state value, $y(\infty) = 1$, allowing their transient responses to be compared directly.

Evaluating Eq. (1) at $z = 1$ gives

$$H(1) = K \frac{1 - z_0}{1 - 2r \cos(\theta) + r^2}.$$

Thus, the required gain is

$$K = \frac{1 - 2r \cos(\theta) + r^2}{1 - z_0}. \quad (7)$$

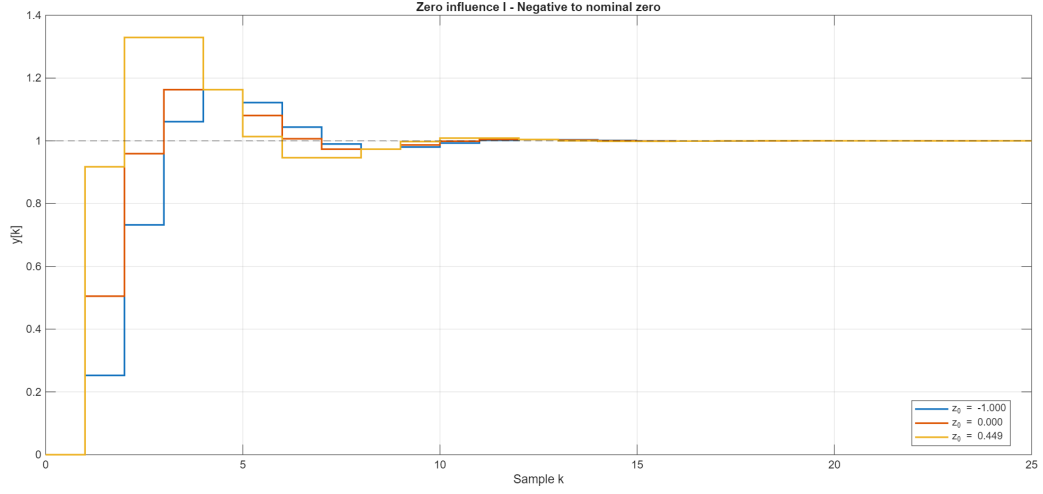
Hence, K itself is not kept constant; it is recalculated so that the steady-state gain remains equal to one.

2 Influence of the Zero Location

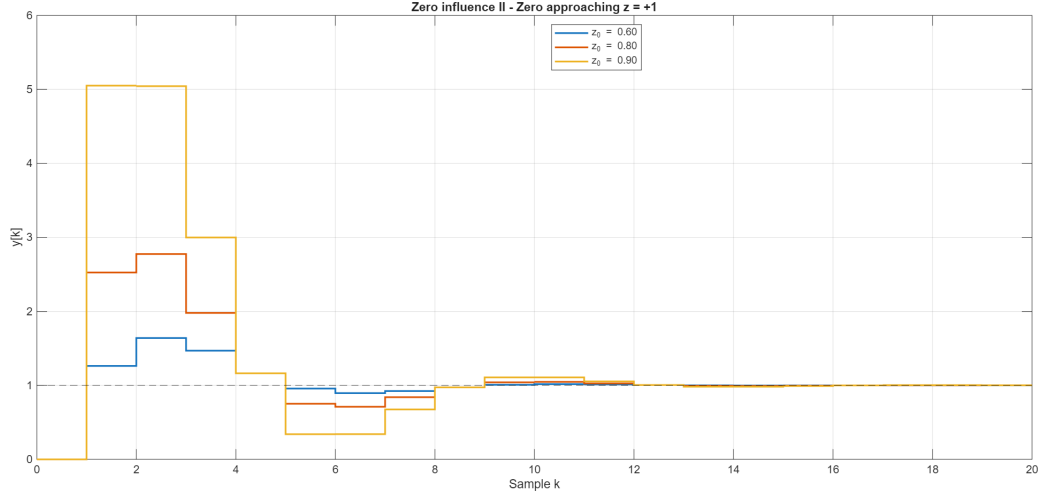
To isolate the effect of the zero, the nominal poles were kept fixed and only z_0 was changed. The tested values were divided into three representative groups:

$$[-1, 0, 0.449], \quad [0.60, 0.80, 0.90], \quad [1.05, 1.10, 1.20].$$

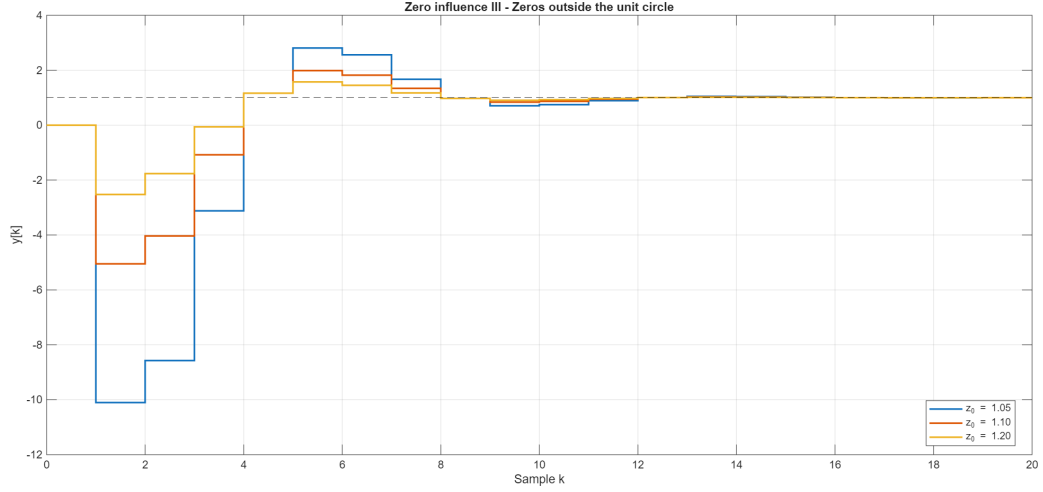
The first group represents zeros on the negative real axis, near the origin, and at the nominal position. The second group illustrates the effect of moving the zero toward $z = 1$, while the third group contains zeros outside the unit circle.



(a) Negative to nominal zero.



(b) Zero approaching $z = 1$.



(c) Zeros outside the unit circle.

Figure 1: Influence of the zero location on the step response.

Figure 1 shows that zeros on the negative real axis or close to the origin have a relatively moderate influence on the response. As the zero approaches $z = 1$, the transient peak increases significantly.

This can partly be understood from Eq. (7). When

$$z_0 \rightarrow 1,$$

the term $1 - z_0$ approaches zero and the gain required to maintain $H(1) = 1$ becomes very large.

The steady-state value therefore remains equal to one, but the transient response can become very large.

For $z_0 > 1$, the zero is outside the unit circle and the system becomes nonminimum phase. The response can initially move in the opposite direction before converging to the final value.

The point $z_0 = 1$ itself cannot be included. In this case,

$$1 - z_0 = 0,$$

and Eq. (7) would require an infinite gain. A zero exactly at $z = 1$ gives zero DC gain for any finite K , making the normalization $H(1) = 1$ impossible.

3 Influence of the Pole Location

For the pole study, the zero was fixed at its nominal value,

$$z_0 = 0.449,$$

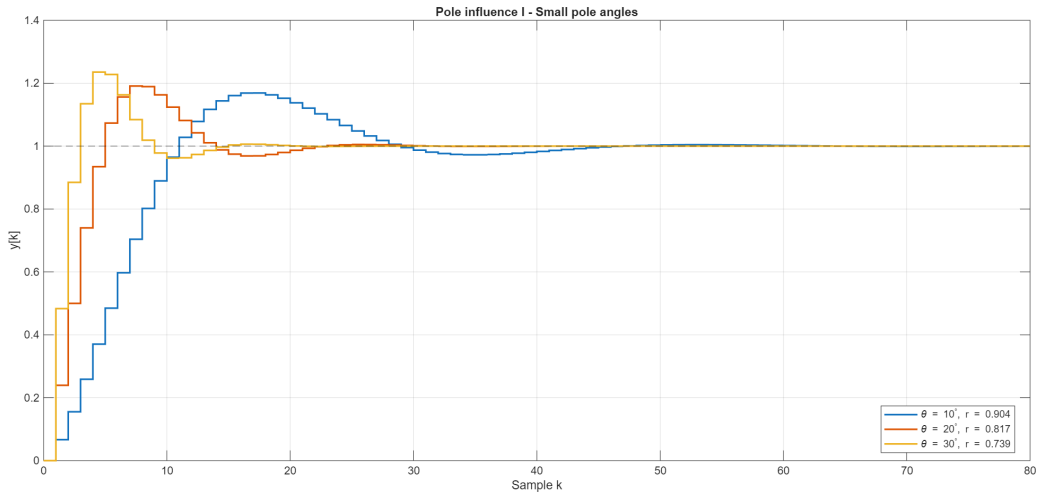
while the complex-conjugate pole pair was moved.

The damping ratio was maintained at

$$\zeta = 0.5.$$

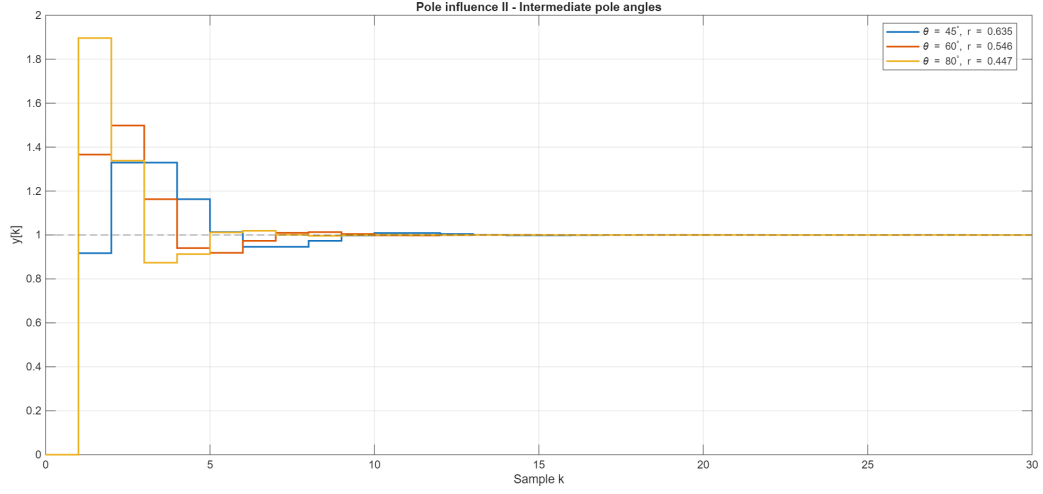
Therefore, each time θ was changed, the corresponding value of r was recalculated using Eq. (5). The tested angles were

$$[10^\circ, 20^\circ, 30^\circ], \quad [45^\circ, 60^\circ, 80^\circ], \quad [100^\circ, 130^\circ, 160^\circ].$$

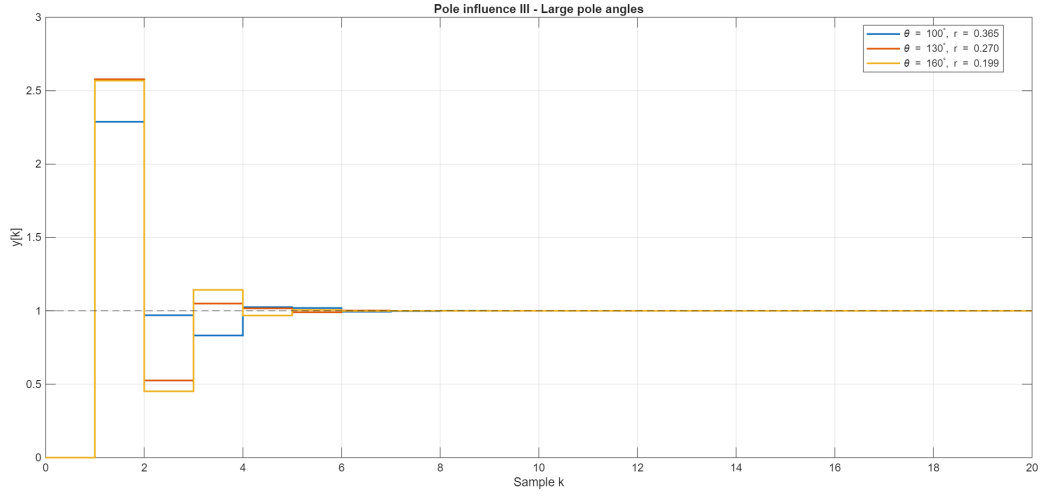


(a) Small pole angles.

Figure 2: Influence of the pole location on the step response for $\zeta = 0.5$.



(b) Intermediate angles.



(c) Large pole angles.

Figure 2: Influence of the pole location on the step response for $\zeta = 0.5$ (continued).

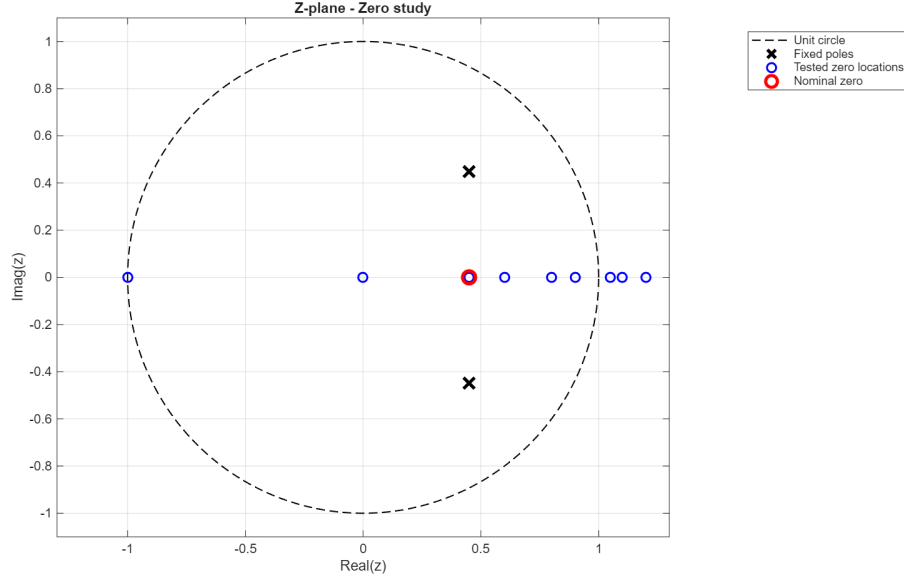
For small values of θ , the poles lie closer to $z = 1$ and relatively close to the unit circle. Their natural modes therefore decay slowly, producing longer responses.

As θ increases, Eq. (5) gives a smaller pole radius. The poles consequently move further inside the unit circle and the response decays more rapidly.

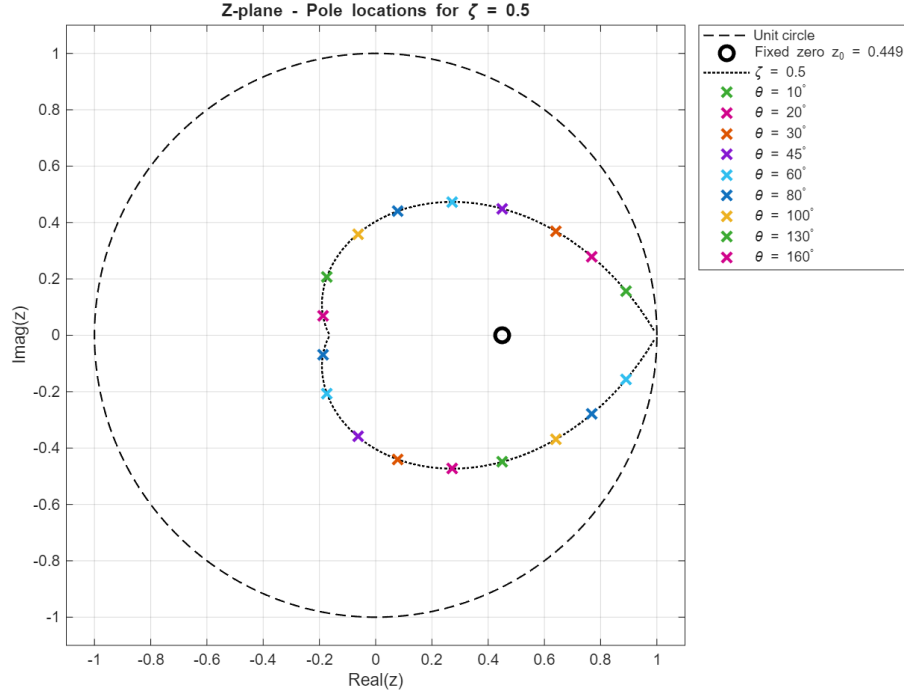
For the largest values of θ , the poles move toward the negative-real side of the z -plane. This also changes the sample-to-sample oscillatory behaviour, although the smaller pole magnitude causes the oscillations to decay rapidly.

4 Pole-Zero Locations in the z -Plane

The tested configurations are summarized in Fig. 3.



(a) Fixed poles and variable zeros.



(b) Fixed zero and variable poles.

Figure 3: Pole-zero configurations used in the simulations.

In the zero study, the poles remain fixed while the zero moves along the real axis. In the pole study, the zero remains fixed and the complex-conjugate poles move along the locus corresponding to $\zeta = 0.5$.

5 Quantitative Analysis

To complement the step-response plots, the percentage overshoot and settling time were calculated for every tested configuration.

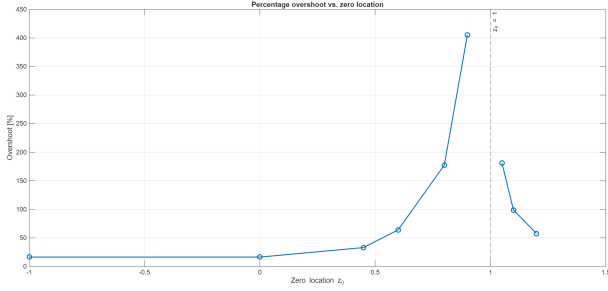
Because all responses have a final value equal to one, the percentage overshoot was calculated as

$$M_p = \max(0, (y_{\max} - 1)100). \quad (8)$$

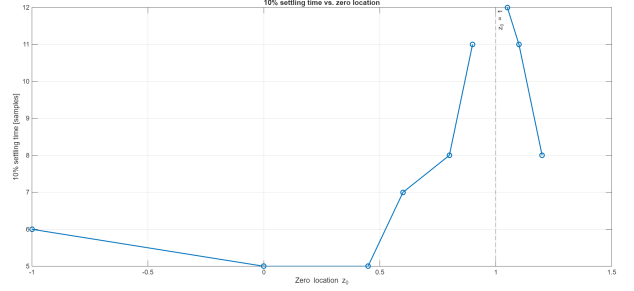
A 10% settling envelope was used. Therefore, the response was considered settled when it entered and permanently remained inside

$$0.9 \leq y[k] \leq 1.1. \quad (9)$$

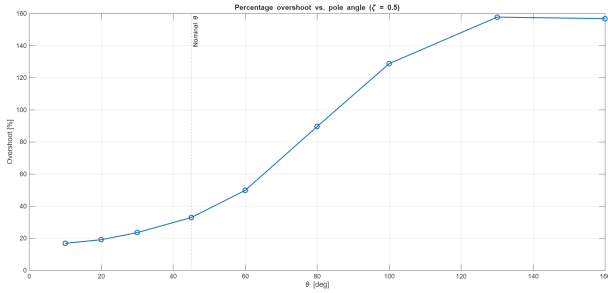
Each marker in Fig. 4 corresponds to one of the systems used in the previous step-response plots. Lines are only used to make the trend easier to visualize.



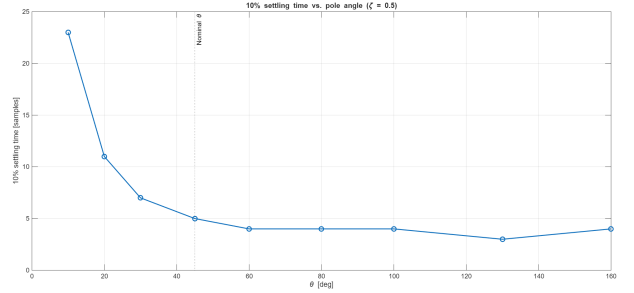
(a) Overshoot vs. zero location.



(b) Settling time vs. zero location.



(c) Overshoot vs. pole angle.



(d) Settling time vs. pole angle.

Figure 4: Quantitative influence of zero and pole location on the transient response.

The zero-location results confirm that the main effect of the zero is on the transient shape and overshoot. In particular, the response becomes increasingly sensitive as the zero approaches $z = 1$. The two branches on either side of $z_0 = 1$ are intentionally not connected, since the unit-gain normalization is undefined at that point.

The pole-location results show a clearer effect on settling time. Small pole angles correspond to poles closer to the unit circle and therefore slower decay. Increasing θ while maintaining $\zeta = 0.5$ decreases the pole radius and generally reduces the settling time.

6 Conclusion

The simulations show that poles and zeros affect the discrete-time step response in different ways. The zero location mainly influences the transient shape and overshoot. Its effect becomes particularly strong close to $z = 1$, while zeros outside the unit circle produce nonminimum-phase behaviour.

The pole location has a more direct influence on the natural response and settling time. Poles closer to the unit circle generate slowly decaying modes, whereas poles further inside the unit circle produce faster responses.

Finally, recalculating K so that $H(1) = 1$ was important to ensure that every system had the same steady-state value. This allowed the influence of pole and zero locations to be compared independently from changes in DC gain.